Learning Diary

Databases and data sources in IoT system

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Foreword

The purpose of this learning diary is to structure the student's experiences with the Databases and data sources and provide feedback to the teacher about the course.

Database

Database is a shared, integrated computer structure that stores a collection of End-user data, or raw facts of interest to the end user**,** metadata, or data about data, through which the end-user data are integrated and managed**.** Database management system (DBMS) is a collection of programs that manages the database structure and controls access to the data stored in the database**.**

There are following types of databases,

- Centralized database.
- Distributed database.
- Personal database.
- End-user database.
- Commercial database.
- NoSQL database.
- Operational database.
- Relational database.
- Cloud database.
- Object-oriented database.
- Graph database.

Relational database

RDBMS always follow structure; several tables' columns and rows are included in this database system. It provides data that can be access in relation to another piece of data in the table. There are millions of rows of data in the table. Data item can be anything, for example, Id, name, age, address, phone, country, images, videos and so on. Relational database is a system that allows to create, update and administer a relational database. The Structured Query Language (SQL) is the standard user and application program interface for a relational database.

Data models

The main function is to help us understand the complexities of the real-world environment. A data model represents, for example, data structures, data characteristics, relationships, constraints and transformations. Data models can facilitate interaction among the designer, the applications programmer and the end user. Data model basic building blocks are Entity, Attribute, Relationship and Constraint.

An entity is anything (a person, a place, a thing or an event) about which data are to be collected and stored, for examples, customer, product, flight route or musical concert.

An attribute is a characteristic of an entity, such as, **Customer** is an entity and it's attributes are Customer last name, Customer first name, Customer phone, Customer address and Customer credit limit.

A relationship describes a association among entities for example: relationship between customer and agent, An agent can serve many customers, and each customer may be served by one agent. Three types of relationships:

- One-to-many (1:*) relationship. Example, Painter (1) paints many different paintings (*).
- Many-to-many (*:*) relationship. Example, An employee may learn many job skills, and each job skill may be learned by many employees.
- One-to-one (1:1) relationship. Example, each store manager, who is a employee, manages only a single store.

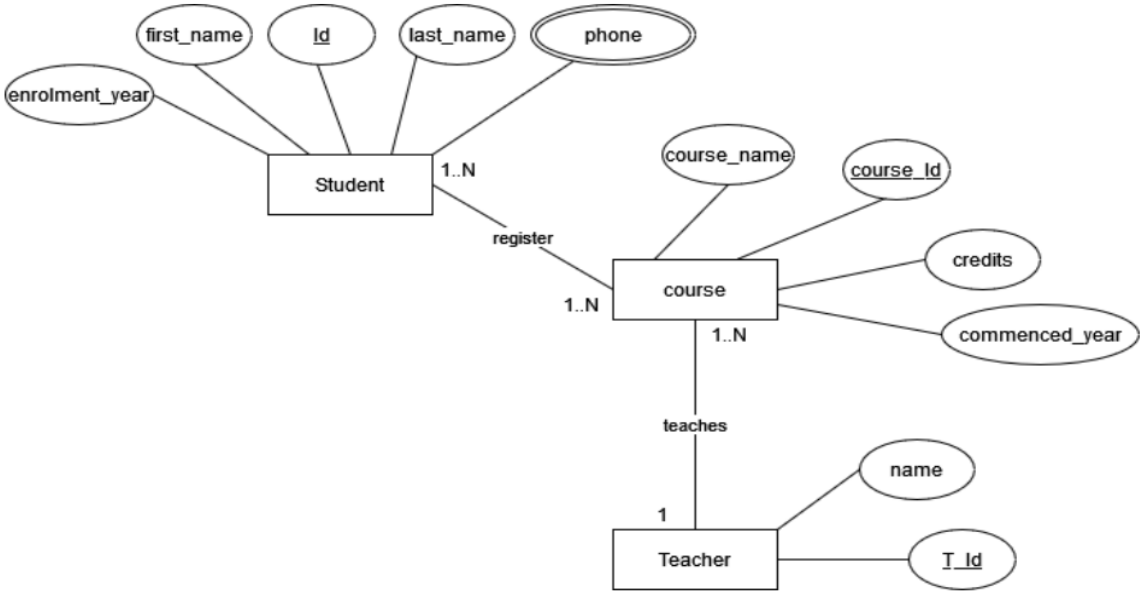
A constraint is a restriction placed on data, such as, salary must have values that are between 6 000 and 350 000 euros or each class must have one and only one teacher.

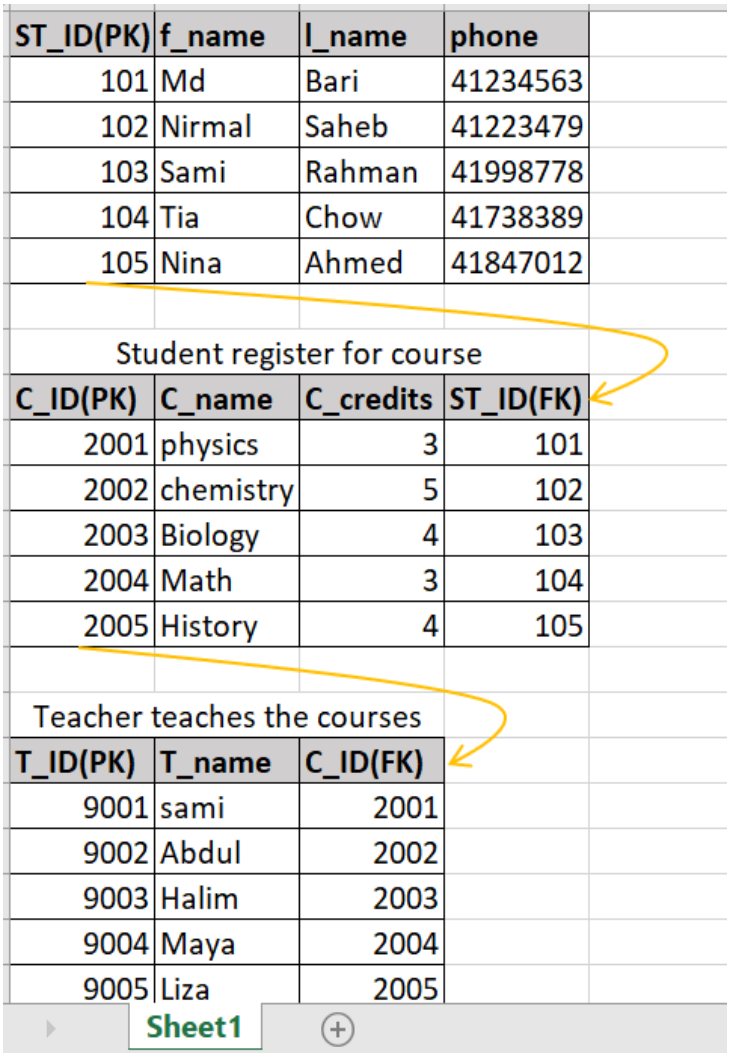
Entity Relationship (ER) model/diagram

Graphical representation of entities and their relationships in a database structure that quickly became popular because it complemented the relational database model concepts. Class diagram component of the Unified Modelling Language (UML) has been used to produce entity relationship models. ER components are Entity is represented as rectangle also entity box and Relationships is represented as line which connects related entities together.



Task.1 Planning the student register





ST_ID(PK)	f_name	l_name	phone
101	Md	Bari	41234563
102	Nirmal	Saheb	41223479
103	Sami	Rahman	41998778
104	Tia	Chow	41738389
105	Nina	Ahmed	41847012

Student register for course

C_ID(PK)	C_name	C_credits	ST_ID(FK)
2001	physics	3	101
2002	chemistry	5	102
2003	Biology	4	103
2004	Math	3	104
2005	History	4	105

Teacher teaches the courses

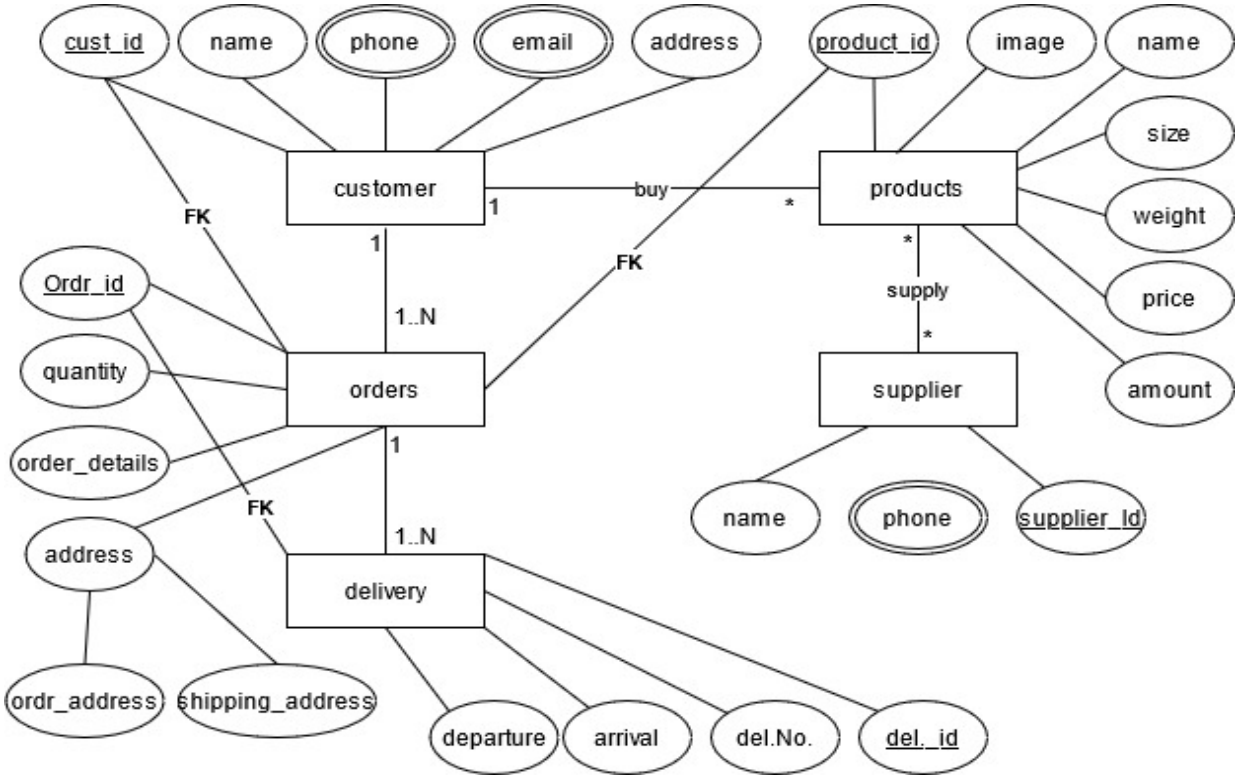
T_ID(PK)	T_name	C_ID(FK)
9001	sami	2001
9002	Abdul	2002
9003	Halim	2003
9004	Maya	2004
9005	Liza	2005

Sheet1

ER diagram for student register and example

Spent almost 8 hours for this exercise(link) because this was first exercise and had to draw diagram. Without understanding, it is very difficult to draw ER diagram. While I was drawing, learned about entity, attribute and relationship and I also have been known about different ERD symbols and notations.

Task.3 Online store database Design



Customer_ID(PK)	f_name	l_name	phone	email	address
101	Md	Bari	41234563	lb@gmai	pori
102	Nirmal	Saheb	41223479	bk@gma	turku
103	Sami	Rahman	41998778	wr@gma	tampere
104	Tia	Chow	41738389	sf@gmai	oulu
105	Nina	Ahmed	41847012	fb@gma	helsinki

Product_ID(PK)	name	price	weight	image	supplier_ID(FK	supplier_name	phone
6001	Camera	50£	300g	image1	7001	Bari	41234563
6002	USB charger	10£	25g	image2	7002	Saheb	41223479
6003	Headset	30£	150g	image3	7003	Rahman	41998778
6004	Color pen	2£	80g	image4	7004	Chow	41738389
6005	Tube Light	8£	50g	image5	7005	Ahmed	41847012

Order_ID(PK)	quantity	details	address	Custome	Product_ID(FK)
101	1	Camera	pori	101	6001
102	2	USB charger	turku	102	6002
103	2	Headset	tampere	103	6003
104	1	Color pen	oulu	104	6004
105	3	Tube Light	helsinki	105	6005

Delivery_ID(PK)	departure	arrival	address	Order_ID(FK)
900101	01/04/2021	10/04/2021	pori	101
900102	02/04/2021	12/04/2021	turku	102
900103	02/04/2021	11/04/2021	tampere	103
900104	01/04/2021	10/04/2021	oulu	104
900105	03/04/2021	15/04/2021	helsinki	105

ER diagram for online store database design and example

Spending time was almost 6 hours for the exercise(link) And learned about keys, such as, primary key, foreign key and composite key. Another thing I learned is that Rows are called Cardinalities and columns are called degree, for example, there are 6 degrees and 5 cardinalities in first table of the online store database.

MySQL

It is a relational database management system based on **SQL** – **S**tructured **Q**uery **L**anguage. The application is used for a wide range of purposes, including data warehousing, e-commerce, and logging applications.

The most common use for MySQL however, is for the purpose of a web database. It can be used to store anything from a single record of information to an entire inventory of available products for an online store.

Mariadb

MariaDB is a community-based MySQL-based relational database system licensed under the GNU GPL. It is developed by the same community as the original MySQL. It is also being developed to have the best possible compatibility with MySQL.

It can be download and install to computer from this link. Installation and configuration link for windows.

Useful MySQL/MariaDB queries CLI

 Log in as root (password is prompted):

```
mysql -u root -p
```

 Log in as root with password `sekrit` (**REMARK**: no space allowed between `-p` and the password):

```
mysql -u root -psekrit
```

 Creating a new user

```
CREATE USER 'monty'@'localhost' IDENTIFIED BY 'some_pass';
```

 Grant permissions to selected databases

```
GRANT ALL PRIVILEGES ON foo.* TO 'monty'@'localhost' WITH GRANT OPTION;
```

 Changing the password

```
SET PASSWORD FOR 'monty'@'localhost' = PASSWORD('uusisalasana');
```

 Creating,deleting,changing a database

```
CREATE/DROP/ALTER DATABASE jokutietokanta;
```

```
USE database_name
```

 Let's see what databases exist on the server?

```
SHOW DATABASES;
```

 Drop user

```
DROP USER 'monty'@'localhost';
```

 Or more simply, if the previous one doesn't work.

```
DROP USER monty;
```

 Another way of assigning rights to a specific database.

```
CREATE USER 'TestUser'@'%' IDENTIFIED BY 'test_paswd';
```

 TO export sql file from mariadb server

```
mysqldump -u username -p database_name > filename.sql
```

 TO import sql file in mariadb server

```
mysql -u username -p database_name < backup.sql
```

The DESCRIBE or DESC command displays the structure of a single table, that is, what fields it contains and what the data type of the field is. Such as `DESCRIBE/DESC tablename;`

Add the user to the database:

```
create user 'eero'@'%' identified by 'pass';
```

All rights are granted to everything:

```
grant all privileges on *.* to 'eero'@'%' with grant option;
```

`User@%` would allow access from all locations. `User@localhost` would only allow access from localhost. They are two different users with two different passwords (though you can set them to the same password, but if you update one password the other will not auto-update)

What is Index in mysql

Indexes are used to find rows with specific column values quickly. Without an index, MySQL must begin with the first row and then read through the entire table to find the relevant rows. The larger the table, the more this costs. If the table has an index for the columns in question, MySQL can quickly determine the position to seek to in the middle of the data file without having to look at all the data. This is much faster than reading every row sequentially.

MySQL Triggers

A MySQL **trigger** is a stored program (with queries) which is executed automatically to respond to a specific event such as insertion, updation or deletion occurring in a table.

1. Before Trigger - Before Trigger fires before the execution of INSERT, DELETE and UPDATE commands.

2. After Trigger - After trigger fires after the execution of INSERT, DELETE and UPDATE commands.

There are 6 different types of triggers in MySQL:

- Before Update Trigger
- After Update Trigger
- Before Insert Trigger
- After Insert Trigger
- Before Delete Trigger
- After Delete Trigger

What is DELIMITER in MySql

The DELIMITER statement changes the standard delimiter which is semicolon (;) to another. The delimiter is changed from the semicolon(;) to double-slashes // or $$.

Why do we have to change the delimiter?

Because we want to pass the stored procedure, custom functions etc. to the server as a whole rather than letting mysql tool to interpret each statement at a time.

MySql Statement

Browsing

```
SHOW DATABASES;
SHOW TABLES;
SHOW FIELDS FROM table / DESCRIBE table;
SHOW CREATE TABLE table;
SHOW PROCESSLIST;
```

Select

```
SELECT * FROM table;
SELECT * FROM table1, table2;
SELECT field1, field2 FROM table1, table2;
SELECT ... FROM ... WHERE condition
SELECT ... FROM ... WHERE condition GROUPBY field;
SELECT ... FROM ... WHERE condition GROUPBY field HAVING condition2;
SELECT ... FROM ... WHERE condition ORDER BY field1, field2;
SELECT ... FROM ... WHERE condition ORDER BY field1, field2 DESC;
SELECT ... FROM ... WHERE condition LIMIT 10;
SELECT DISTINCT field1 FROM ...
SELECT DISTINCT field1, field2 FROM ...
```

Select - Join

```
SELECT ... FROM t1 JOIN t2 ON t1.id1 = t2.id2 WHERE condition;
SELECT ... FROM t1 LEFT JOIN t2 ON t1.id1 = t2.id2 WHERE condition;
SELECT ... FROM t1 JOIN (t2 JOIN t3 ON ...) ON ...
```

Insert

```
INSERT INTO table1 (field1, field2) VALUES (value1, value2);
```

Update

```
UPDATE table1 SET field1=new_value1 WHERE condition;
UPDATE table1, table2 SET field1=new_value1, field2=new_value2, ...
WHERE table1.id1 = table2.id2 AND condition;
```

Delete

```
DELETE FROM table1 / TRUNCATE table1
DELETE FROM table1 WHERE condition
DELETE FROM table1, table2 FROM table1, table2 WHERE table1.id1 =
  table2.id2 AND condition
```

Create / Delete / Modify Table

Create

```
CREATE TABLE table (field1 type1, field2 type2);
CREATE TABLE table (field1 type1, field2 type2, INDEX (field));
CREATE TABLE table (field1 type1, field2 type2, PRIMARY KEY
(field1));
CREATE TABLE table (field1 type1, field2 type2, PRIMARY KEY
(field1,field2)
```

DROP

```
DROP TABLE table;
DROP TABLE IF EXISTS table;
DROP TABLE table1, table2, ...
```

ALTER

```
ALTER TABLE table MODIFY field1 type1
ALTER TABLE table MODIFY field1 type1 NOT NULL ...
ALTER TABLE table CHANGE old_name_field1 new_name_field1 type1
ALTER TABLE table CHANGE old name field1 new name field1 type1 NOT
```